

Torus 1-point functions for the $\mathfrak{sl}(2)$ WZW models and their modular properties

Simon Wood
School of Mathematics
Cardiff University

Joint work with
Matthew Krauel and Jamal Shafiq
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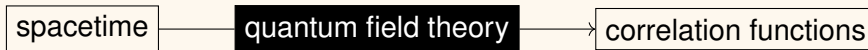
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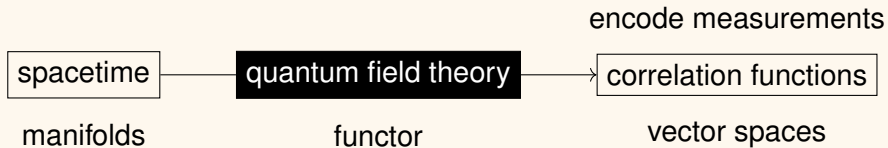
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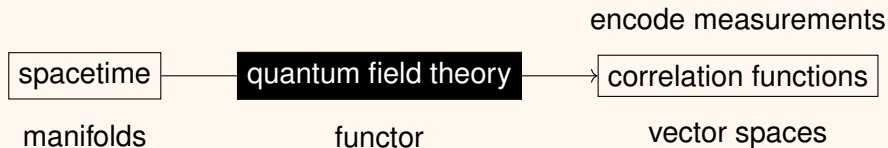
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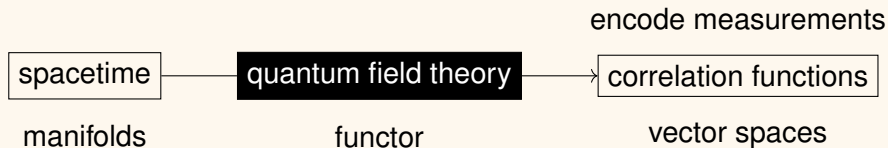
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For 2-dimensional conformally invariant quantum field theories (CFTs) the spacetimes can be

- oriented with Euclidean conformal structure (equivalent to a complex structure),
 - chiral CFT on a complex curve
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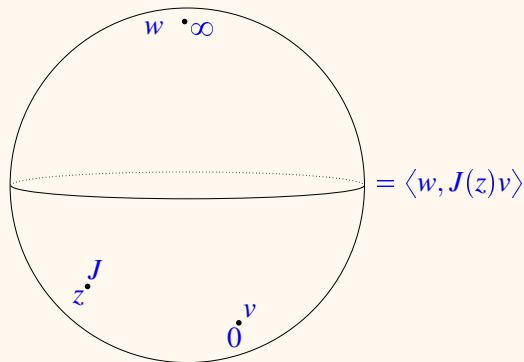
Full local CFTs can be built by glueing together two chiral ones (holomorphic + antiholomorphic = real analytic).

Motivation: The gist of vertex algebras

A vertex algebra V is an algebra of “currents”

$$J(z) = \sum_{n \in \mathbb{Z}} J_n z^{-n-1}, \quad J_n \in \text{End}(\mathcal{M}).$$

The variable z is to be understood as a coordinate on the Riemann sphere $\mathbb{P}$.

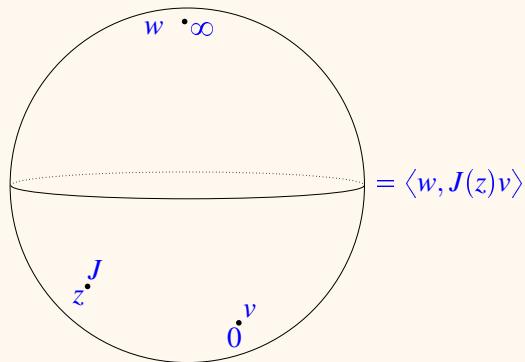


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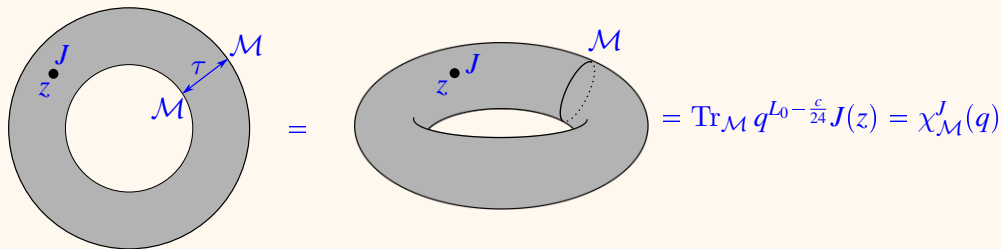


This constructs correlation functions on the Riemann sphere.
But a CFT should be well defined on any Riemann surface!

$$J \in V, v \in \mathcal{M}, w \in \mathcal{M}^*, \mathcal{M} \in V\text{-Mod}.$$

From 3 points on $\mathbb{P}$ to 1 point on the torus, $\mathbb{T}$

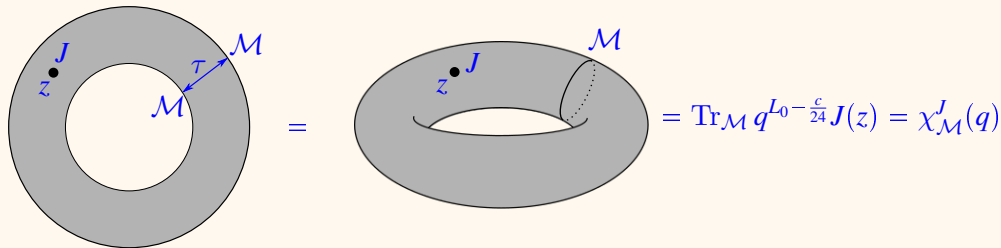
Cut out and glue an annulus in $\mathbb{P}$:



Here, $q = \exp(2\pi i \tau)$ and $\mathcal{M} \in V\text{-Mod}$.

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For $J = \mathbf{1}$ (vacuum), we get the character (graded dimensions) of $\mathcal{M}$.

Complex structures on the torus

Many different $q = \exp(2\pi i\tau)$ give tori with equivalent complex structures.
The symmetry groups (mapping class groups) are

empty torus: $\mathrm{PSL}(2, \mathbb{Z}) = \langle S^2 = (ST)^3, S^2 = 1 \rangle, T(\tau) = \tau + 1, S(\tau) = -\frac{1}{\tau}$

punctured torus: $B_3 = \langle S^2 = (ST)^3 \rangle$

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- B_3 actions in CFT are projective actions for $\mathrm{SL}(2, \mathbb{Z})$ and can be “deprojectified” by suitable cocycles.

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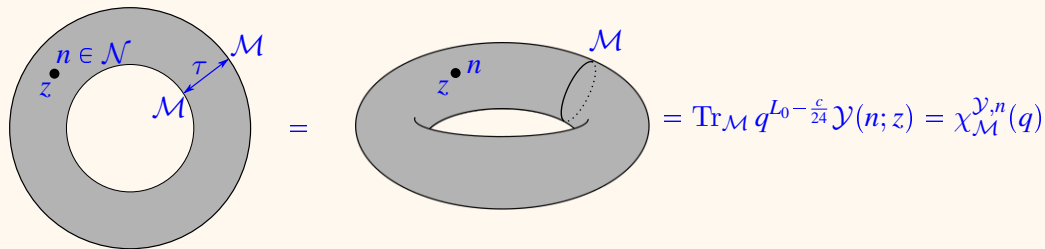
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$$\mathcal{C}^1(V) = \mathrm{span}\{\chi_{\mathcal{M}}^\bullet : \mathcal{M} \in V\text{-Mod}\} \hookrightarrow \mathrm{SL}(2, \mathbb{Z}), B_3$$

1-point functions should allow insertions from any module!

Again cut out and glue an annulus in $\mathbb{P}$, but with the insertion in a module $\mathcal{N}$:



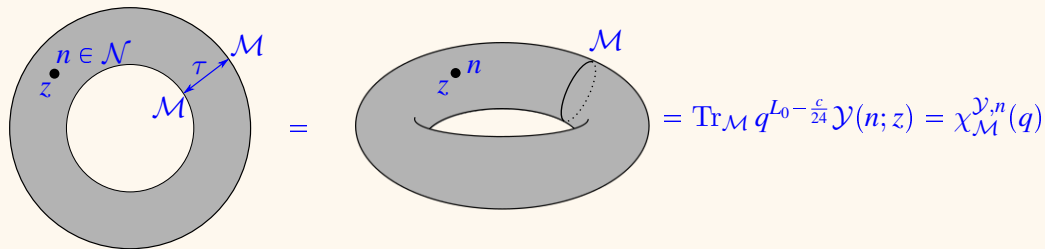
$$= \text{Tr}_{\mathcal{M}} q^{L_0 - \frac{c}{24}} \mathcal{Y}(n; z) = \chi_{\mathcal{M}}^{\mathcal{Y}, n}(q)$$

Here, $\mathcal{Y}$ is an intertwining operator (a V -bilinear map).

This can only be non-zero if $\mathcal{N}$ acts on $\mathcal{M}$, i.e. if $\mathcal{N} \boxtimes \mathcal{M} \twoheadrightarrow \mathcal{M}$.

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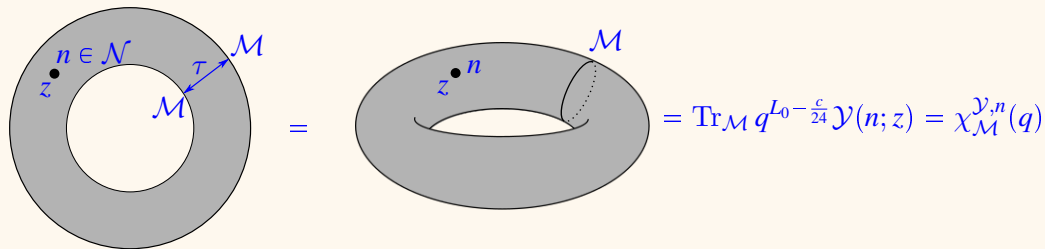
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Challenge: Can we determine $\mathcal{C}^1(\mathcal{N})$?

Understanding $\mathcal{C}^1(\mathcal{N})$ from (modular) tensor categories

Theorem

Explicit formulae for the B_3 generators from categorical data.

$$S^\lambda: \begin{array}{c} \mu \\ | \\ \bullet a \\ / \quad \backslash \\ \lambda \quad \mu \end{array} \mapsto \sum_{\nu, b} S^\lambda_{\mu, a; \nu, b} \cdot \begin{array}{c} \nu \\ | \\ \bullet b \\ / \quad \backslash \\ \lambda \quad \nu \end{array}, \quad S^\lambda_{\mu, a; \nu, b} = \frac{1}{S_{0,0}} \cdot \begin{array}{c} \mu \quad \nu \\ \curvearrowright \quad \curvearrowright \\ a \quad \bar{b} \\ \lambda \end{array}$$

Understanding $\mathcal{C}^1(\mathcal{N})$ from analytic number theory

Pick a basis $\{f_1, \dots, f_\ell\}$ of

$$\mathcal{C}_n^1(\mathcal{N}) = \text{span}\{\chi_{\mathcal{M}}^{\mathcal{Y},n}(q) : \chi_{\mathcal{M}}^{\mathcal{Y}}(q) \in \mathcal{C}^1(\mathcal{N})\}$$

then $\psi_{\mathcal{N}} = (f_1, \dots, f_\ell)^t$ is a vector valued modular form (vvmf) whose weight is the conformal weight of n :

- $f_i : \mathbb{H} \rightarrow \mathbb{C}$ holomorphic with potential pole at $i\infty$.
- For $\gamma = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in \text{SL}(2, \mathbb{Z})$

$$f_i|_{\gamma} = \nu(\gamma)^{-1} (c\tau + d)^{-\text{wt } n} f_i \left(\frac{a\tau + b}{c\tau + d} \right),$$

where ν is a multiplier system (converts B_3 actions into $\text{SL}(2, \mathbb{Z})$ actions).

- $\psi_{\mathcal{N}}|_{\gamma} = \rho(\gamma)\psi_{\mathcal{N}}$, where $\rho : \text{SL}(2, \mathbb{Z}) \rightarrow \text{GL}(\ell, \mathbb{C})$ is a representation.

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For $\mathcal{N} \neq V$, usually $\text{wt } n \notin \mathbb{Z}$, which is highly **non-standard**.

Vector valued modular forms are well studied in analytic number theory

Common questions:

- Is $\text{im}(\rho)$ finite? Congruence?
- Let $M(\rho, k) = \text{span}\{\text{all vvmfs with rep } \rho, \text{ weight } k\}$. Is $M(\rho, k)$ finitely generated as a module over the rings of differential modular forms R or modular functions M ?
- In the context here: How big is $\mathcal{C}_n^1(\mathcal{N})$ in $M(\rho, \text{wt } n)$?
- In the context here: Is there information loss when going $\mathcal{C}^1(\mathcal{N}) \twoheadrightarrow \mathcal{C}_n^1(\mathcal{N})$?

Example: $\mathfrak{sl}(2)$ at level $k \in \mathbb{N}_0$

- Generating currents: e, h, f with $[h, e] = 2e, [h, f] = -2f, [e, f] = h$

Operator product relation: $x(z)y(w) \sim \frac{k\langle x, y \rangle}{(z-w)^2} + \frac{[x, y](w)}{z-w} + \text{reg}$

Normal ordering relation: $: e(z)^{k+1} := 0$

- Modules: simple highest weight $\mathfrak{sl}(2)$ -modules

$L(k, \lambda), 0 \leq \lambda \leq k$

- Fusion rules: truncation of Clebsch-Gordan

$$\dim(\lambda \boxtimes \mu \rightarrow \nu) = \begin{cases} 1 & \lambda + \mu + \nu \in 2\mathbb{Z}, |\lambda - \mu| \leq \nu \leq \min\{\lambda + \mu, 2k - \lambda - \mu\} \\ 0 & \text{otherwise} \end{cases}$$

- only even λ can “act” and only on μ satisfying $\frac{\lambda}{2} \leq \mu \leq k - \frac{\lambda}{2}$

- For λ even: $\dim \mathcal{C}^1(\lambda) = \sum_{\mu=\frac{\lambda}{2}}^{k-\frac{\lambda}{2}} \dim(\lambda \boxtimes \mu \rightarrow \mu) = k - \lambda + 1$

Specialising from $\mathcal{C}^1(\lambda)$ to $\mathcal{C}_u^1(\lambda)$, $u \in L(k, \lambda)$

Question:

Can we find $u \in L(k, \lambda)$ such that

$$\mathrm{Tr}_{L(k, \mu)} \left[q^{L_0 - \frac{c}{24}} u_0 \right] = \chi_\mu^u(q) \neq 0?$$

Here, u_0 is the zero mode of the current $u(z) = \sum_{n \in \mathbb{Z}} u_n z^{-n - \mathrm{wt} u}$.

- If the $\mathfrak{sl}(2)$ weight of u is not 0, then $\chi_\mu^u(q) = 0$.
- First “sensible” candidate $u = f_0^{\frac{\lambda}{2}} |\lambda\rangle$. However,

$$\left(f_0^{\frac{\lambda}{2}} |\lambda\rangle \right)_0 = \left[f_0, \left(f_0^{\frac{\lambda}{2} - 1} |\lambda\rangle \right)_0 \right]$$

and traces of commutators vanish because traces are cyclic.

Non-zero torus 1-point functions

Lemma

If $u \in L(k, \lambda)$ generates a simple non-trivial finite $\mathfrak{sl}(2)$ module, then $\chi_\lambda^u(q) = 0$.

Theorem

- 1 The trivial finite $\mathfrak{sl}(2)$ module appears for the first time in $L(k, \lambda)$ at degree $\frac{\lambda}{2}$. The multiplicity at this degree is 1.
- 2 The vector spanning this trivial module has the form $u = f_{-1}^{\frac{\lambda}{2}} |\lambda\rangle + \text{corrections}$.
- 3 $\chi_\mu^u(q) \neq 0$
- 4 The set $\{\chi_\mu^u(q) : \frac{\lambda}{2} \leq \mu \leq k - \frac{\lambda}{2}\}$ is linearly independent.
- 5 $\Psi_\lambda(u, q) = (\chi_{\frac{\lambda}{2}}^u(q), \dots, \chi_{k-\frac{\lambda}{2}}^u(q))^t$ is a vvmf.
- 6 $\dim \mathcal{C}_u^1(\lambda) = \dim \mathcal{C}^1(\lambda) = k - \lambda + 1$, so no information loss from specialising to u .

$\dim \mathcal{C}^1(\lambda) = 1$, thus $\lambda = k$, k even

Let ρ_λ be the representation of $\mathrm{SL}(2, \mathbb{Z})$ obtained from $L(k, \lambda)$ and $\mathcal{V}(\rho_\lambda) = \mathrm{span}\{\Psi_\lambda(w, q) : w \in L(k, \lambda)\}$.

Theorem

- 1 The representation ρ_k is irreducible, finite image and congruence.
- 2 $\rho_k(\mathbf{S}) = e^{2\pi i \frac{k}{8}}$, $\rho_k(\mathbf{T}) = e^{2\pi i \frac{k}{24}}$. Trivial if $k \in 24\mathbb{Z}$.
- 3 $\Psi_k(u, q) = \eta^{\frac{3k}{2}}$
- 4 $\mathcal{V}(\rho_k) = R\eta^{\frac{3k}{2}}$
- 5 All vvmfs in $\mathcal{V}(\rho_k)$ are holomorphic (no poles at $i\infty$).
- 6 As an M module, $\mathcal{V}(\rho_k)$ is free and rank 1 with basis $\{\eta^{\frac{3k}{2}}\}$ for $k \geq 2$.
- 7 Graded dimensions of $\mathcal{V}(\rho_k)$ known with an explicit formula.

$\dim \mathcal{C}^1(\lambda) = 2$, thus $\lambda = k - 1$, k odd

Let

$$\Phi = \begin{pmatrix} J^{\frac{1}{24}} {}_2F_1\left(\frac{-1}{24}, \frac{7}{24}; \frac{3}{4}; J^{-1}\right) \\ J^{-\frac{5}{24}} {}_2F_1\left(\frac{5}{24}, \frac{13}{24}; \frac{5}{4}; J^{-1}\right) \end{pmatrix}$$

Theorem

- 1 The representation ρ_{k-1} is irreducible, finite image and congruence.
- 2 $\rho_{k-1}(\mathbf{T}) = \text{diag}\left(\mathbf{e}^{2\pi i \frac{k-2}{24}}, \mathbf{e}^{2\pi i \frac{k+4}{24}}\right)$
- 3 $\Psi_{k-1}(u, q) = \eta^{\frac{3k^2+2k-5}{2(k+2)}} \Phi$
- 4 $\mathcal{V}(\rho_{k-1}) = R\Psi_{k-1}(u, q)$.
- 5 All vvmfs in $\mathcal{V}(\rho_{k-1})$ are holomorphic for $k \geq 3$.
- 6 As an M module, $\mathcal{V}(\rho_{k-1})$ is free and rank 2 with basis $\{\Psi_{k-1}(u, q), \partial\Psi_{k-1}(u, q)\}$ for $k \geq 2$.
- 7 Graded dimensions of $\mathcal{V}(\rho_{k-1})$ known with an explicit formula.

Level k	Cyclic generator $\Psi_{k-1}(u, q)$
3	$q^{3/40} \left(1 + \frac{1}{5}q - \frac{117}{25}q^2 - \frac{84}{125}q^3 + \frac{3659}{625}q^4 + \dots \right)$ $q^{13/40} \left(1 - \frac{9}{5}q - \frac{2}{25}q^2 - \frac{39}{125}q^3 - \frac{126}{625}q^4 + \dots \right)$
5	$q^{11/56} \left(1 - \frac{19}{7}q - \frac{264}{49}q^2 + \frac{6061}{343}q^3 + \frac{22963}{2401}q^4 + \dots \right)$ $q^{25/56} \left(1 - \frac{33}{7}q + \frac{247}{49}q^2 + \frac{1672}{343}q^3 - \frac{18183}{2401}q^4 + \dots \right)$
7	$q^{23/72} \left(1 - \frac{17}{3}q + \frac{23}{9}q^2 + \frac{3128}{81}q^3 - \frac{13429}{243}q^4 + \dots \right)$ $q^{41/72} \left(1 - \frac{23}{3}q + \frac{170}{9}q^2 - \frac{391}{81}q^3 - \frac{10948}{243}q^4 + \dots \right)$
9	$q^{39/88} \left(1 - \frac{95}{11}q + \frac{2340}{121}q^2 + \frac{48165}{1331}q^3 - \frac{2895523}{14641}q^4 + \dots \right)$ $q^{61/88} \left(1 - \frac{117}{11}q + \frac{5035}{121}q^2 - \frac{74100}{1331}q^3 - \frac{1011465}{14641}q^4 + \dots \right)$
11	$q^{59/104} \left(1 - \frac{151}{13}q + \frac{7611}{169}q^2 - \frac{35636}{2197}q^3 - \frac{9959957}{28561}q^4 + \dots \right)$ $q^{85/104} \left(1 - \frac{177}{13}q + \frac{12382}{169}q^2 - \frac{383087}{2197}q^3 + \frac{1229442}{28561}q^4 + \dots \right)$
13	$q^{83/120} \left(1 - \frac{73}{5}q + \frac{1992}{25}q^2 - \frac{18177}{125}q^3 - \frac{224261}{625}q^4 + \dots \right)$ $q^{113/120} \left(1 - \frac{83}{5}q + \frac{2847}{25}q^2 - \frac{48472}{125}q^3 + \frac{309009}{625}q^4 + \dots \right)$

$\dim \mathcal{C}^1(\lambda) = 3$, thus $\lambda = k - 2$, k even

Let

$$\Phi_k = \begin{pmatrix} J^{\frac{(k+1)}{12(k+2)}} {}_3F_2 \left(-\frac{(k+1)}{12(k+2)}, \frac{11k+14}{24(k+2)}, \frac{19k+30}{24(k+2)}; \frac{3k+7}{4(k+2)}, \frac{1}{2}; J^{-1} \right) \\ J^{-\frac{k+1}{6(k+2)}} {}_3F_2 \left(\frac{k+1}{6(k+2)}, \frac{3k+5}{6(k+2)}, \frac{5k+9}{6(k+2)}; \frac{5k+9}{4(k+2)}, \frac{5}{8}; J^{-1} \right) \\ J^{-\frac{5k+11}{12(k+2)}} {}_3F_2 \left(\frac{5k+11}{12(k+2)}, \frac{9k+19}{12(k+2)}, \frac{13k+27}{12(k+2)}; \frac{3}{2}, \frac{5k+11}{4(k+2)}; J^{-1} \right) \end{pmatrix}.$$

Theorem

- ① ρ_{k-2} is irreducible and finite image.
- ② Order of $\rho_{k-2}(\mathbb{T}) = 12(k+2)$ if $k \in 4 + 6\mathbb{Z}$ or $4(k+2)$ otherwise. ρ_{k-2} is non-congruence if $o(\rho_{k-2}(\mathbb{T})) \nmid 25,401,600 = 2^8 \cdot 3^4 \cdot 5^2 \cdot 7^2$.
- ③ $\Psi_{k-2}(u, q) = \eta^{\frac{3k^2-2k-8}{2(k+2)}} \Phi_k$ and $\mathcal{V}(\rho_{k-2}) = R\Psi_{k-2}(u, q)$.
- ④ All vvmfs in $\mathcal{V}(\rho_{k-2})$ are holomorphic for $k \geq 4$.
- ⑤ As an M module, $\mathcal{V}(\rho_{k-2})$ is free and rank 3 with basis $\{\Psi_{k-2}(u, q), \partial\Psi_{k-2}(u, q), \partial^2\Psi_{k-2}(u, q)\}$ for $k \geq 4$.
- ⑥ Graded dimensions of $\mathcal{V}(\rho_{k-2})$ known with an explicit formula.

Level k	$\Psi_{k-2}(u, q)$
4	$q^{1/24} \left(1 - \frac{6991}{171}q - \frac{1462930981}{198531}q^2 - \frac{11520966474250}{5360337}q^3 - \frac{467661528323716250}{627159429}q^4 + \dots \right)$ $q^{1/4} \left(1 + \frac{134}{9}q + \frac{167509}{81}q^2 + \frac{24672291010}{45927}q^3 + \frac{2054193740460070}{11986947}q^4 + \dots \right)$ $q^{13/24} \left(1 - \frac{31}{27}q + \frac{473}{1215}q^2 - \frac{27056}{32805}q^3 - \frac{1533931}{2657205}q^4 + \dots \right)$
6	$q^{5/32} \left(1 - \frac{1041}{20}q - \frac{28822341}{3040}q^2 - \frac{34699584029}{12160}q^3 - \frac{2170275413391777}{2140160}q^4 + \dots \right)$ $q^{3/8} \left(1 + \frac{74}{5}q + \frac{317943}{130}q^2 + \frac{8423595}{13}q^3 + \frac{21692516271}{104}q^4 + \dots \right)$ $q^{21/32} \left(1 - \frac{31}{8}q + \frac{423}{128}q^2 + \frac{14247}{7168}q^3 - \frac{485683}{229376}q^4 + \dots \right)$
8	$q^{11/40} \left(1 - \frac{46803}{775}q - \frac{14944931541}{1375625}q^2 - \frac{574656427747084}{171953125}q^3 - \frac{782261040133149248781}{649123046875}q^4 + \dots \right)$ $q^{1/2} \left(1 + \frac{1704}{125}q + \frac{108483138}{40625}q^2 + \frac{5094872662288}{7109375}q^3 + \frac{5893213005533601}{25390625}q^4 + \dots \right)$ $q^{31/40} \left(1 - \frac{503}{75}q + \frac{44149}{3125}q^2 - \frac{206842}{78125}q^3 - \frac{420276376}{17578125}q^4 + \dots \right)$
10	$q^{19/48} \left(1 - \frac{44717}{666}q - \frac{14421863479}{1222776}q^2 - \frac{243672512766437}{66029904}q^3 - \frac{68038738170466662661}{50617747584}q^4 + \dots \right)$ $q^{5/8} \left(1 + \frac{107}{9}q + \frac{2963152}{1053}q^2 + \frac{64959522367}{85293}q^3 + \frac{5516615806491181}{22261473}q^4 + \dots \right)$ $q^{43/48} \left(1 - \frac{259}{27}q + \frac{8110}{243}q^2 - \frac{251140}{6561}q^3 - \frac{25036652}{531441}q^4 + \dots \right)$

$$\dim \mathcal{C}^1(\lambda) \geq 4$$

Theorem

Let the level be $k = p^t - 2$, where $p > 3$ is prime and t is a positive integer. For $2 \leq \lambda \leq k$ with λ even, if the representation ρ_λ is irreducible, then it is non-congruence if $t = 1$ or if $t > 1$ and $\lambda + 1 > p^{t-2}$.

Proposition (using the categorical formulae)

Choose $p = 7$, $t = 1$ above and hence $k = 5$. Then for $\lambda = 2$, the representation ρ_2 is 4-dimensional, irreducible and non-congruence.

Theorem

Let $\lambda = k - 3$, k odd. If $\mathcal{H}(\rho_{k-3}, \nu_{\frac{-k^2+4k+6}{2(k+2)}})$ (the space of holomorphic vvmfs) is a cyclic R -module, then

$$\mathcal{V}^u(\rho_\lambda) = \mathcal{V}(\rho_\lambda) = \eta^{\frac{3}{2} \frac{k^2-4k-3}{k+2}} \mathcal{H}\left(\rho_{k-3}, \nu_{\frac{-k^2+4k+6}{2(k+2)}}\right).$$

Outlook

- Analytic number theory tools very helpful up to dimension 3. Some results might generalise up to dimension 5. Bad behaviour expected in higher dimensions.
- Need to find better ways to combine number theory and category theory tools.
- Category theory data well known for $\mathfrak{sl}(2)$ but higher ranks look hard.
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Thank you!